College Placement Management System

Complete Website Development Requirements

1. System Overview

The system shall be a cloud-based web application used to manage the complete college placement process. It shall provide separate access for Students, Placement Officers/TPOs, Recruiters/Companies, and System Administrators.

The website shall be responsive and usable on desktop, tablet, and mobile screens.

2. User Roles

- Student: maintains profile, resume and academic information; views eligible jobs; applies; tracks recruitment stages and results.
- Placement Officer/TPO: manages students, companies, drives, eligibility, applications, schedules, results, notifications and reports.
- Recruiter/Company: manages company profile, jobs, applicants, shortlisting, recruitment rounds and results.
- System Administrator: manages users, roles, permissions, system settings and system-level monitoring.

3. Website Pages

- Landing/Home page
- Login page
- Registration page
- Forgot/Reset Password page
- Student Dashboard
- Student Profile
- Resume Management
- Placement Drives / Jobs
- Placement Drive Details
- My Applications
- Application Details / Recruitment Status
- Test and Interview Schedule
- Notifications
- Student Placement Result
- TPO Dashboard
- Student Management
- Company Management
- Placement Drive Management
- Application and Shortlist Management
- Recruitment Round Management
- Schedule Management
- Result Management
- Notification Management
- Reports and Statistics
- Recruiter Dashboard
- Recruiter Company Profile

- Job/Drive Management
- Applicant Management
- Shortlist and Recruitment Rounds
- Recruitment Result
- Admin Dashboard
- User and Role Management
- System Settings and Audit Logs

4. Common Website Requirements

- Header shall show the website name/logo, current user name, notifications and logout option.
- Sidebar/navigation shall show only the pages permitted for the logged-in user's role.
- Every data list shall provide search, filtering, sorting and pagination where required.
- Forms shall show required fields, validation errors and success/error messages.
- Delete operations shall require confirmation.
- The system shall display loading indicators during server operations.
- The system shall handle empty states, invalid requests and server errors clearly.
- The interface shall use consistent buttons, labels, tables, forms, cards and status indicators.

5. Authentication Requirements

- User shall log in using registered credentials.
- System shall identify the user's role after login and open the correct dashboard.
- Passwords shall never be stored as plain text.
- Forgot-password and reset-password functionality shall be provided.
- Inactive or blocked accounts shall not be allowed to log in.
- Logout shall invalidate the active session/token.
- Protected pages shall not be accessible without authentication.
- Users shall not be able to access functions belonging to another role.

6. Student Module

6.1 Student Registration

- Fields: student ID/roll number, full name, email, mobile number, password, department/branch and graduation year.
- Student ID and email shall be unique.
- Required fields shall be validated before registration.
- Email and mobile number shall be validated.
- Registration shall create a student account with the Student role.

6.2 Student Profile

- Personal details: name, email, phone, date of birth, gender and address.
- Academic details: 10th percentage, 12th/diploma percentage, CGPA, backlogs, branch and graduation year.
- Skills: technical and soft skills.
- Education: degree/course and department.
- Certifications, internships and project details.
- Profile shall support editing and saving.
- System shall show profile completion status if implemented.

6.3 Resume

- Student shall upload a resume.
- Allowed file type and maximum file size shall be defined by the administrator.
- Student shall view, replace and delete the current resume.
- Resume shall be stored in cloud storage; database shall store its secure reference.

6.4 Student Dashboard

- Show profile completion.
- Show number of available/eligible placement drives.
- Show applied drives.
- Show upcoming tests/interviews.
- Show latest notifications.
- Show current placement status.

6.5 Placement Drives

- Student shall see active placement drives.
- Each drive shall show company, role, package/CTC, location, deadline, eligibility and selection process.
- Student shall be able to open a detailed drive page.
- Apply button shall be available only when the student is eligible and the application period is open.
- System shall prevent duplicate applications.

6.6 Application Status

- Student shall view all submitted applications.
- Status values shall include at least Applied, Shortlisted, Rejected, Test Scheduled, Interview Scheduled, Selected and Waitlisted.
- Student shall see the recruitment stage and relevant schedule information.

7. Placement Officer/TPO Module

7.1 TPO Dashboard

- Show total students, registered students, companies, active drives, applications and placed students.
- Show placement percentage and useful placement statistics.
- Show upcoming drives and recruitment activities.
- Show recent notifications and important actions.

7.2 Student Management

- View all students.
- Search by name, student ID, department, graduation year or other relevant fields.
- View complete student profile and resume.
- Edit student information when authorized.
- Activate/deactivate student accounts.
- Import student records from CSV/Excel if required.
- Export student data when authorized.

7.3 Company Management

- Add, edit, view and deactivate companies.
- Fields: company name, industry, description, website, contact person, email and phone.

- Manage recruiter accounts linked to companies.
- View company placement history.

7.4 Placement Drive Management

- Create, edit, publish, close and archive placement drives.
- Fields: company, job title, description, location, salary/CTC, employment type, application start/end date, drive date and selection process.
- Define eligible branches, graduation year, minimum CGPA, academic percentage requirements, maximum backlogs and required skills.
- Drive status shall include Draft, Published, Closed and Archived.

7.5 Eligibility

- System shall automatically compare student data against drive eligibility rules.
- Eligibility rules shall support CGPA, percentages, backlogs, branch, graduation year and other configured criteria.
- TPO shall be able to view eligible and non-eligible students.
- The system shall not allow an ineligible student to apply unless an authorized override feature is explicitly enabled.

7.6 Application and Shortlisting

- TPO shall view applications for each drive.
- TPO shall filter applicants by eligibility, department and application status.
- TPO/recruiter shall be able to shortlist or reject applicants.
- Status changes shall be stored with date/time and responsible user.
- Students shall be notified when appropriate.

7.7 Recruitment Rounds

- Create rounds such as Aptitude Test, Coding Test, Technical Interview, HR Interview and Final Selection.
- Assign students to a round.
- Record Pass, Fail, Pending or other configured outcomes.
- Move successful students to the next round.

7.8 Scheduling

- Create test/interview schedules.
- Fields: round, date, time, venue/online link, instructions and participating students.
- Prevent conflicting schedules where applicable.
- Notify participating students.

7.9 Results

- Record final selection status.
- Store selected company, job role, package/CTC, selection date and joining date when available.
- Allow authorized users to correct results.
- Maintain placement history.

7.10 Reports

- Student placement report.
- Company-wise placement report.
- Department-wise placement report.
- Year-wise placement report.
- Drive/application report.

- Highest, lowest and average package where salary data is available.
- Total placed students and placement percentage.
- Reports shall support export where required.

8. Recruiter/Company Module

- Recruiter shall log in using a company-linked account.
- Recruiter shall manage company profile subject to TPO permissions.
- Recruiter shall create or request job/placement drives depending on the college workflow.
- Recruiter shall define job role, description, location, package and eligibility criteria.
- Recruiter shall view eligible applicants.
- Recruiter shall shortlist/reject applicants when authorized.
- Recruiter shall manage recruitment rounds and interview/test outcomes.
- Recruiter shall provide final selection results to the college.
- Recruiter shall not access unrelated student data or other companies' data.

9. Notification Module

- Create and display in-app notifications.
- Notify students about new drives, deadlines, shortlisting, schedules and results.
- Notify recruiters/TPOs about relevant applications and recruitment activities.
- Store notification read/unread status.
- Email notification can be integrated for important events.

10. Database Requirements

- Users
- Students
- Departments
- Skills
- Student Skills
- Companies
- Recruiters
- Placement Drives
- Eligibility Criteria
- Applications
- Recruitment Rounds
- Round Results
- Schedules
- Notifications
- Resumes/Documents
- Placement Records
- Audit Logs

All tables shall use appropriate primary keys, foreign keys, indexes, constraints and timestamps. Relationships shall prevent orphan records and invalid references.

11. Important Validation Requirements

- Required fields shall not accept empty values.
- Email shall have valid format.
- Phone number shall have valid length/format.
- CGPA and percentages shall be within valid ranges.
- Backlogs shall not be negative.
- Application deadline shall be valid and consistent with the drive dates.
- Student shall not apply twice to the same drive.
- Only eligible students shall normally be allowed to apply.
- Uploaded files shall be checked for type and size.
- Unauthorized users shall receive an access-denied response.
- Duplicate company, recruiter, student ID and email records shall be prevented where applicable.

12. Security Requirements

- Use HTTPS in production.
- Hash passwords using a secure password-hashing algorithm.
- Implement role-based access control on both frontend and backend.
- Validate and sanitize user input.
- Use parameterized queries/ORM to prevent SQL injection.
- Protect against unauthorized file access.
- Apply appropriate session/token expiration.
- Do not expose passwords, tokens or sensitive information in logs.
- Record important administrative actions in audit logs.

13. Cloud Requirements

- Frontend and backend shall be deployable to cloud infrastructure.
- Database shall be hosted using a cloud database service or cloud-hosted database server.
- Resumes/documents shall use cloud object storage.
- Environment variables/secrets shall be used for database credentials and API keys.
- HTTPS/SSL shall be configured for production.
- Automated or scheduled database backup shall be configured.
- Application shall be scalable for increasing students, companies and applications.
- Cloud monitoring/logging shall be enabled where available.
- Application shall have separate development/test and production configuration.

14. API/Backend Requirements

- Backend shall expose authenticated APIs for users, students, companies, drives, applications, schedules, notifications and results.
- APIs shall validate incoming data.
- APIs shall enforce role permissions; frontend restrictions alone are not sufficient.
- APIs shall return consistent success and error responses.
- Sensitive endpoints shall require authentication.
- File upload APIs shall validate file type, size and authorization.

15. Non-Functional Requirements

- Performance: common pages and API operations should respond quickly under expected college workload.
- Availability: application should be accessible through the internet when cloud services are available.
- Scalability: architecture should allow additional students, departments and colleges to be added.
- Reliability: failures should be handled without corrupting placement data.
- Usability: screens should be simple enough for students and staff with basic computer knowledge.
- Responsiveness: website should work on desktop, tablet and mobile.
- Maintainability: code should be modular, documented and version-controlled.
- Backup and recovery: important database and document data shall have recovery procedures.

16. Development Deliverables

- Responsive frontend website.
- Backend/API application.
- Cloud-hosted database.
- Cloud document/resume storage.
- Authentication and role management.
- Database schema and relationships.
- API documentation.
- Deployment configuration.
- Test cases and test results.
- User/admin documentation.